Fiscal Year:	FY 2021	**Task Last Updated:**	FY 09/09/2020
PI Name:	DeChurch, Leslie Ph.D.		
Project Title:	Team Task Switching in Astronaut Crews on the International Space Station: Integrating Multiteam Membership, Multiteam Systems, Multitasking,& Multidimensional Networks to Monitor & Enable Functional Work Shifts in Astronaut Crews		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: Previously at Georgia Institute of Technology until July 2016.		
Project Type:	Flight	**Solicitation / Funding Source:**	2013-14 HERO NNJ13ZSA002N-ILSRA. International Life Sciences Research Announcement
Start Date:	11/09/2017	**End Date:**	11/08/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	6	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Contractor, Noshir Ph.D. (Northwestern University)		
Grant/Contract No.:	80NSSC18K0276		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Team Task Switching in Astronaut Crews on the International Space Station: Integrating Multiteam Membership, Multiteam Systems, Multitasking, & Multidimensional Networks to Monitor & Enable Functional Work Shifts in Astronaut Crews," grant NNX15AK73G, due to Principal Investigator move to Northwestern University from Georgia Tech in Fall 2016. We are at the dawn of a new era of human space exploration. Moving beyond low Earth orbit and the relative safety of the International Space Station (ISS) toward near-Earth asteroids and Mars presents previously unimaginable opportunities as well as organizational challenges. One significant challenge is the complexity of the operating environment within which astronauts will work. This complexity will place enormous demands on astronauts, and research is needed that develops concrete countermeasures to mitigate the risks stemming from performance decrements due to inadequate cooperation, coordination, communication, and psychosocial adaptation within a team. Astronauts will push the bounds of human cognitive and social functioning as they multitask across tasks, teams, and tools working toward personal, team, and system goals. This multidisciplinary research project is designed to help them do just that. This investigation into team task switching leverages: (1) agent-based models to understand how task switching behavior and performance-related switching costs occur based on dynamic interplay between independent and interdependent tasks, (2) laboratory experiments conducted in two multiteam systems laboratories (one at Georgia Tech, the other at Northwestern) to test the theoretical model derived from agent-based models, (3) unobtrusive data collection strategies to capture information on the tasks and complex social network structures of ISS crew members, (4) design of interventions to enable adaptive team task switches using virtual experiments, (5) validation of intervention strategies using one of NASA's space analogs, and (6) development of a dashboard decision aid to anticipate and pre-empt dysfunctional task switching. An innovative feature of our investigation is the use of a multidimensional network approach to characterize and model the switches between tasks, tools, teams, and multi-team systems. The combined outputs of this proposed multi-disciplinary project speak directly to NASA Human Research Program's (HRP's) identified team gaps for autonomous, long duration, and/or long distance exploration missions associated with the need to identify (1) key threats to the team (Team Gap 1), (2) countermeasures to support team function (Team Gap 3), (3) validated methods that can be used to maintain team function (Team Gap 5), and (4) psychological and psychosocial factors, measures, and combinations thereof that can be used to compose highly effective crews (Team Gap 8). [Ed. note Sept 2020: HRP Gaps have changed since original proposal--see Human Research Roadmap https://]
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Previous models of task switching revolve around task and individual factors that affect work efficiency as individuals move from task to task throughout their workday. However, individuals not only accomplish tasks individually using one technological tool, they do so as members of multiple teams using a variety of technologies. Our research on team task switching examines factors that affect the decision and ability of individuals to switch between tasks with varying levels of interdependence, as well as discuss the mediators and moderators that determine adaptive switching in today's complicated work environments. This model affords a comprehensive understanding of how individuals adapt to dynamic, environmentally-triggered performance demands requiring them to change tasks, teams, and technologies, and shift back and forth between personal, team, and system goals. Our research will be useful for investigating task switching in any modern-day organization facing complex collaborative challenges, such as NASA space exploration, large scientific consortia (e.g., CERN-the European Organization for Nuclear Research), cybersecurity teams, healthcare systems, and the military. Furthermore, findings could then be leveraged to develop system-wide interventions that increase overall work efficiency and resilience in safety-critical systems.
	Over the course of this five-years programmatic research program, we (1) explore issues that can arise when astronauts switch between tasks, teams, and technologies within the complex, multiteam systems that characterize operations on the International Space Station (ISS) and during Long Duration Space Exploration (LDSE), as well as (2) investigate the effects of astronaut task, team, and technology switches on individual-, team-, and system-level performance. In particular, astronauts work individually as well as within and across different teams to accomplish their work. They also complete a variety of tasks and engage a variety of technologies in the process. Shifting across these tasks, teams, and tools holds the potential for "switch costs" in the form of process loss and performance decrements. Aspects of the various tasks, teams, and technologies make switching to/from each more/less seamless. We explore the implications of effective/efficient switching. In Year 1 we developed a conceptual framework to guide our thinking in this area, and conducted research in Year 2 that allows us to begin to draw conclusions about the role of task/team/technology switches on individual/team/multiteam process and performance. Our conceptual framework highlights three key types of switches (task, team, and technology) as well as two directions of switching (lateral versus vertical). Lateral switches occur when individuals shift across tasks, teams, or technologies/tools. Such lateral shifts can be either singular or compound. A singular shift entails changes in one of the three elements (e.g., the individual switches from working on task A to task B or from working with tool A to tool B). In contrast, a compound shift entails changes on two or more of the task, team, and tool dimensions (e.g., a compound task-team shift means the individual goes from working on task A with individuals X, Y, and Z to working on task B with individuals I, J, and K). Our framework also describes vertical switches, which involve a shift in the level of interdependence (e.g., individual, single team, multiple teams) needed to complete a task. Vertical shifts can be either upward (e.g., working independently to working with a team) or downward (e.g., working with a team to working independently). In Year 2 of our project, we conducted a five-pronged effort at understanding the potential role of team/task/tool switching in long duration space exploration: (1) policy-capturing research to explore the factors that affect attraction to different teams, (2) laboratory studies to explore ability to effectively make switches, (3) agent based model development to explore the psychological, task, tool, and social network based factors that determine desire to stay with or switch from a task/team/tool, (4) analog studies using HERA (Human Exploration Research Analog) crews to explore how task characteristics affected vertical and lateral switches, and (5) a task switching survey aboard the ISS. In Year 3 of our project, we: (1) continued to improve and run virtual experiments using our agent based model, (2) conducted empirical analyses of analog crew task behavior to uncover factors affecting task switching, (3) implemented our task switching survey aboard the ISS, and (4) developed our protocols for studying task switching in the SIRIUS (Scientific International Research In a Unique terrestrial Station) mission located in the Russian NEK (Nezemnyy

Task Progress:	Eksperimental'nyy Kompleks, Russia's Institute of Biomedical Problems ground-based experimental complex). In Year 4 of our project, we: (1) continued to improve and run virtual experiments using our agent based model, (2) conducted empirical analyses of analog crew task behavior to uncover factors affecting task switching, (3) implemented our task switching survey aboard the ISS, (4) conducted semi-structured interviews with 5 astronauts returning from the ISS, and (4) conducted our study in the Russian NEK SIRIUS study. In Year 5 of our project, we: (1) continued to conduct empirical analyses of analog crew task behavior to uncover factors affecting task switching, (2) analyzed the task switching survey we performed aboard the ISS, (3) provided recommendations for designing work in space based on our computational model and survey we performed aboard the ISS, and (4) developed measures of multiteam spaceflight performance. Our research will provide fundamental, generalizable findings in the area of efficient task switching across teams and multiteam systems. Specifically, previous models of task switching revolve around task and individual factors that affect work efficiency as individuals move from task to task throughout their workday. However, individuals not only accomplish tasks individually using one technological tool, they do so as members of multiple teams using a variety of technologies. Our conceptual framework of team task switching lays out a framework of vertical and lateral shifts, the sources of inertia, and the mediators and moderators that determine adaptive switching in today's complicated work environments. This model affords a comprehensive understanding of how individuals adapt to dynamic, environmentally-triggered performance demands requiring them to change tasks, teams, and technologies, and shift back and forth between personal, team, and system goals. Our framework will be useful for investigating task switching in any modern-day organization facing complex collaborative challenges, such as NASA space exploration, large scientific consortia (e.g., CERN), cybersecurity teams, healthcare systems, and the military. Furthermore, findings could then be leveraged to develop system-wide interventions that increase overall work efficiency and resilience in safety-critical systems. In this review period, we have collected data across multiple platforms (HERA and SIRIUS) to begin to understand the effects of variables across our conceptual framework on individual and team task switching performance, preferences, and motivations. In addition to providing Earth benefits in the area of efficient task switching across teams, tasks, and technologies, we are also testing our hypotheses in analogs that recreate the unique contexts in which astronauts operate (i.e., extreme, isolated/confined environments). We will compare results across research paradigms to see how effects of isolation/confinement may moderate team task switching behavior and performance, thus providing unique findings that will inform NASA's operations aboard the International Space Station (ISS) and on Long-Duration Space Exploration (LDSE) missions.
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